✓ Assessing DNSSEC vulnerability to AI swarms

Yes, but the picture is more nuanced than "low DNSSEC = wide open." The DNSSEC gap is real, but the more pressing AI-era DNS threats actually **bypass DNSSEC entirely** by exploiting trust assumptions rather than cryptographic weaknesses.

Where DNSSEC matters for AI agents:
The ~99.5% of unvalidated queries are vulnerable to spoofing and poisoning. AI agents that resolve domains autonomously and at scale have no human in the loop to notice a redirect, so a successful DNS spoof against an agent is silently fatal. AI can also optimize the *timing and prediction* of spoofing attempts (transaction IDs, response windows), raising success rates against the unvalidated majority.

Where DNSSEC is irrelevant (and the bigger risk):

- **Dangling DNS / subdomain takeover** — Akamai called this "the most overlooked attack surface in the AI era." A forgotten CNAME to a decommissioned cloud resource gets reclaimed by an attacker, who now controls a *trusted* subdomain. AI agents follow it without question and can be fed hidden prompt-injection payloads via SVG metadata or invisible HTML. No DNSSEC signature is needed — the record is legitimately signed by the owner.
- **DNS rebinding** — a TOCTOU gap where the IP changes between validation and connection. Almost no AI agent platform defends against this.
- **AI-driven domain generation** — ML DGAs produce names statistically indistinguishable from legitimate traffic, defeating entropy-based detection. Infoblox's 2025 report found 100.8M new domains in a year, 25% malicious.

Agent swarms specifically:
Cisco Talos reported in October 2026 that AI agent swarms are *already* breaching networks, with DNS/C2 beaconing as a primary detection target. The recent CyberXero incident (51 agents exfiltrating 628K records in 61 seconds) shows the operational reality. Attackers are also using AI to *defend* their DNS infrastructure — rotating domains, matching legitimate query timing, blending malicious traffic.

Ask a follow-up question

AI-generated answer. Please verify critical facts.